

DAILY LOOP

```
capsule status → capsule env show → capsule config customer show → capsule list --users → capsule term <tag> → work → capsule cleanup
```

01 – AUTH & IDENTITY

```
capsule auth login          # browser
capsule status              # identity
+ expiry
capsule auth logout
capsule auth storage        # OneDrive OAuth
capsule auth token          # print token
```

Browser flow falls back to manual token at oxmiq.ai/oxcapsule/auth. Headless: export **CAPSULE_AUTH_TOKEN**.

02 – ENVIRONMENT

```
prod  mihirainfra.b2c · internal prod
public oxcapsule.b2c · external/partner
dev   dev / test
demo  demonstrations
capsule env show
capsule env set public
capsule auth login          # always
re-auth
```

public & the others use **different B2C tenants** — separate user registries. Switching env requires fresh login.

03 – CUSTOMER FLEET

```
micc  default main fleet
modelhosting model hosting GPUs
oneplay OnePlay partner machines
cree8  Cree8 partner machines
capsule config customer show
capsule config customer set
modelhosting
capsule config customer unset #
env default
```

If **capsule list** looks wrong, this is the **FIRST** thing to check.

04 – LISTING & FILTERING

```
capsule list                # by config tag
capsule list --all          # unique IDs
capsule list --users        # who is on
capsule list --json | jq .
--filter "vendor=nvidia,vram>=24"
keys  vendor, gpu, os, vram, memory, cores, ci
vendors nvidia, tenstorrent, amd, intel
ci= true / false
```

PowerShell: use **capsule** (not cap) and quote the filter — > is redirection in PS.

05 – CONNECT (ONE ROUTING LANGUAGE)

```
capsule term remote shell (default SshRTC)
capsule exec one-off command
capsule code VS Code Remote-SSH
capsule cursor Cursor Remote-SSH
capsule claude Claude Code Desktop
```

Shared flags: **<tag>** or **-u <id>**, **--direct**, **--idle-timeout**, **--max-session-length**, **--turn.code/cursor** also accept **--repo owner/repo**. **term** aliases: **terminal**, **ssh**, **launch**.

06 – CONNECTION RECIPES

```
capsule term <tag>
capsule term -u boostergold461
capsule term <tag> -e "nvidia-smi"

# port-forward (direct SSH)
capsule term <tag> --direct \
  --options "-L 3000:localhost:3000"

# bounded session
capsule term <tag> \
  --idle-timeout 2h \
  --max-session-length 8h

# IDE with repo clone
capsule code <tag> --repo owner/repo
```

07 – FILES & STORAGE

```
capsule auth storage        # once, OAuth
ls ~/OneDrive               # auto-mounted

# SCP
capsule scp upload <tag> ./src /dst
capsule scp download <tag> /src ./dst + --direct, --options "-l 50000"

# Dotfile passthrough
capsule config files add .gitconfig
~/.gitconfig
capsule config files list
capsule config files remove .vimrc
capsule config files clear

# Custom mount
capsule config storage-mount-dir set /workspace/cloud
capsule config storage-mount-dir unset
```

08 – STREAM & DOCKER

```
capsule stream <tag>          #
full desktop
capsule stream <tag> --app "blender"

capsule docker <tag>          #
ubuntu:latest
capsule docker <tag> \
  --image pytorch/pytorch:latest \
  --memory 16
capsule docker <tag> \
  --image nvidia/cuda:12.0-devel \
  --command "python train.py" \
  --volume "/data:/data:ro;/workspace:/workspace"
capsule docker <tag> -- --gpus all # passthrough
```

Hardware encoding: **NVENC** / **VAAPI** / **VideoToolbox**. Adaptive bitrate; clipboard sync both ways.

09 – COMFYUI & CLAUDE MCP

```
capsule comfy <tag>          #
default :8188
capsule comfy <tag> --port 8190 \
  --data-dir ~/my-comfyui \
  --hf-token hf_xxx
capsule comfy <tag> --image my/comfyui:custom

# Open in browser at http://localhost:8188
```

```
# Capsule MCP for Claude Desktop/Code
capsule mcp                      #
install
capsule mcp --uninstall
capsule mcp --output ./mcp.json #
inspect
```

10 – CAPSULE AGENT (PRE-GA)

```
export CAPSULE_AGENT_ENABLED=1
export OXMIQ_AGENT_API_BASE=http://127.0.0.1:8080/v1
export OXMIQ_AGENT_API_KEY=your-api-key
export OXMIQ_AGENT_MODEL=openai/capsule

capsule
agent                      #
chat
capsule agent -p "list nvidia >=16GB VRAM" # one-shot
capsule agent -c "show me free GPUs"    # iterative
```

Requires **uv** installed (brew install uv) and a valid OpenAI-compatible endpoint.

TRIAGE LOOP – WHEN SOMETHING IS BROKEN

reproduce → capture: --version, env, customer, cmd, timestamp, unique-id → capsule cleanup → retry → --direct fallback → escalate with logs

11 – BENCHMARK · CORE FLAGS

```
capsule benchmark <tag> <model>
-c / --concurrency batch (default 4)
--isl / --osl in / out tokens (128 / 128)
-n / --num-prompts total reqs (c × 10)
--tp tensor-parallelism (1)
-q / --quant awq, gptq, fp8, int8, Q4_K_M, Q5_K_M
-b / --backend vllm, llamacpp, mlx, oxpython
--api-base/-key external endpoint, no deploy
--no-upload skip dashboard upload
--idle/--max timeouts; always set for long runs
```

12 – BENCHMARK RECIPES

```
# Standard 8B eval
capsule benchmark <tag> \
  meta-llama/Llama-3.1-8B-Instruct \
  -c 8 --isl 256 --osl 256 -n 40

# Multi-GPU
capsule benchmark <tag> <model> --tp 4

# Quantized comparison
capsule benchmark <tag> <model> --quant awq
capsule benchmark <tag> <model> \
  --backend llamacpp --quant Q4_K_M

# Against existing endpoint
capsule benchmark --api-base http://localhost:8000 \
  --api-key test <model>

# Long-run safety
capsule benchmark <tag> <model> \
  --idle-timeout 4h --max-session-length 8h
```

13 – CHAT FLAGS

```
capsule chat <tag> <model>
-b / --backend override auto-detect
--temperature sampling (default 0.7)
--max-tokens output cap
--system-prompt prepended each turn
--agent aichat agent + MCP tools (oxsol)
--hf-token for gated HF models
--rm cleanup container on exit
--no-stream disable streaming
--client-name aichat client id (default capsule)
--api-base/-key talk to existing server
```

14 – SCHEDULE (BATCH JOBS)

```
capsule schedule start <tag> \
  --script ./train.sh --name "v2" \
  --timeout 8h --retry 3 \
  --env "EPOCHS=100" --env "LR=0.001" \
  --with-file ./model.py

# Target a named machine
capsule schedule start --machine-name NV-4090-01 \
  --script ./benchmark.sh

# Monitor
capsule schedule status <tag> --me \
  --state running --show-start
capsule schedule status <tag> --user x@example.com
capsule schedule logs <job-id> --tail 100

# Cancel
capsule schedule cancel <job-id>
capsule schedule cancel --all

Use for anything > 1 hour. Don't tie up GPUs on idle terminals.
```

15 – TRIAGE CHEAT CODES

Wrong machines env show + customer show
SshRTC fails cleanup → retry → --direct
VS Code remote err rm capsule-* in ~/.ssh/ config
macOS keychain "Always Allow" once
PS filter > use capsule, quote arg
Update fails close SshRTC sessions first
gh download 401 re-check GH_TOKEN scopes
Auth fails (browser) manual token at oxmiq.ai/oxcapsule/auth
~/OneDrive empty re-run capsule auth storage
Benchmark vanished check timeouts; logs; dashboard

16 – DIAGNOSTICS

```
capsule status # auth + expiry
capsule env show # current env
capsule config customer show # current fleet
capsule cleanup # tear down stale state
capsule --version # build id (for bugs)
capsule --no-banner list # clean output
capsule list --users # who else is on
capsule list --json | jq . # machine-readable

# Update
capsule update --check-only
capsule update
capsule update --pre-release
```

17 – ENV VARS

GH_TOKEN install / update (scopes: repo, read:org, workflow, user)
CAPSULE_AUTH_TOKEN headless auth
CAPSULE_SSH_PRIVATE_KEY override SSH key
CAPSULE_SSH_PUBLIC_KEY override pub key
CAPSULE_DEFAULT_APP_CONFIG default app config
CAPSULE_AGENT_ENABLED enable agent (pre-GA)
HF_TOKEN gated HF models
OXMIQ_AGENT_API_BASE agent upstream URL
OXMIQ_AGENT_API_KEY agent API key
OXMIQ_AGENT_MODEL agent model name
HOMEbrew_GITHUB_API_TOKEN macOS install (=GH_TOKEN)

18 – ALIASES & IDIOMS

cap same as capsule (PS quirks with >)
terminal / ssh / launch aliases of term
<tag> pool name → any available
-u <id> specific physical machine
--direct bypass WebRTC, use TCP SSH
--turn url explicit WebRTC relay
--idle-timeout kill if idle for X
--max-session-length hard cap on session
--repo owner/r code & cursor only — clones on connect

19 – CONFIG FILES & PATHS

macOS ~/Library/Application Support/Capsule/
Windows %APPDATA%\capsule\
Linux ~/.config/capsule/
capsule.conf main configuration
capsule-customer.conf customer override
capsule_rsa(.pub) auto-generated SSH key
rcclone.conf cloud storage config
config-files.json file passthrough mapping
capsule config banner show / hide
capsule config bashrc-override enable / disable

20 – RESOURCES

Bench dashboard oxcapsulebenchmark.z22.web.core.windows.net
Manual token oxmiq.ai/oxcapsule/auth
Install repo mihira-ai/ox.capsule
Brew tap mihira-ai/software-packages
Pre-release brew install capsule-pre
Update CLI capsule update [--pre-release]
Linux PATH ~/.local/capsule-cli/bin
Windows PATH C:\Program Files\Capsule-CL\bin
Discord help #capsule-help